lit-acquisition: A Multi-source Multilingual Biomedical Literature Acquisition System with Citation Graph Augmentation

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ABSTRACT

Variant classification in clinical genetics (e.g., ACMG/AMP guidelines) requires clinicians to assemble evidence from biomedical literature that is scattered across many databases, publishers, preprint servers, and languages. Manual acquisition is slow and leaves coverage gaps, especially for non-English literature. We present **lit-acquisition**, an open-source pipeline that integrates 18+ heterogeneous literature providers — PubMed, EuropePMC, Crossref, OpenAlex, Semantic Scholar, preprint servers, language-specific databases (CiNii, J-STAGE, SciELO), and others — behind a single acquisition interface. The system contributes four ideas: (i) a unified acquisition pipeline with health-aware parallel search and schema normalization; (ii) LLM-based cross-lingual query translation into six languages with language-aware provider routing; (iii) citation graph traversal over Semantic Scholar to discover relevant work beyond keyword search; and (iv) an LLM relevance gate and open-access-aware download layer that retrieves full texts only from legal OA sources.

We evaluate the system on a gold standard of 30 clinically relevant genetic variants spanning 541 annotated papers. An ablation study (K=50) shows that the full pipeline improves Recall@50 from 0.171 (PubMed-only) to 0.348 (+0.177, Wilcoxon $p < 0.001$), with each component — multi-source search, cross-lingual translation, and citation graph traversal — contributing additively. Citation graph traversal alone contributes 11 relevant papers that no keyword method finds, concentrated in high-yield variants such as EGFR and DMPK. Multilingual search recovers non-English (German, French, Chinese) relevant literature that English-only search misses. The toolkit and benchmark are released open-source for reproducible evaluation.

CCS CONCEPTS

• **Information systems** → **Information retrieval**; *Search engine indexing*; • **Computing methodologies** → *Natural language processing*.

KEYWORDS

biomedical literature, multilingual retrieval, citation graph, cross-lingual IR, variant classification, ACMG/AMP

1 INTRODUCTION

Interpreting a genetic variant for clinical reporting requires assembling published evidence: case reports, functional studies, sequencing cohorts, and guidelines that establish whether a variant is pathogenic or benign. Under the ACMG/AMP guidelines [1], this evidence supports criteria such as PS3 (well-established functional effect) and PVS1 (null variant). Evidence is scattered across publisher databases (PubMed, Crossref, EuropePMC), preprint servers (arXiv, bioRxiv, medRxiv), regional and language-specific repositories (CiNii, J-STAGE, SciELO), and trial and data registries (ClinicalTrials.gov, Zenodo). Clinicians and genetic counselors routinely search several of these systems manually, deduplicate results by hand, and risk missing relevant non-English literature.

Commercial tools such as Mastermind [5] and systematic-review platforms (Covidence [6], Rayyan [7]) address parts of this workflow, but they are closed-source, cover limited sources, and do not provide automated open-access full-text retrieval with explicit license awareness.

We present **lit-acquisition**, an open-source, MIT-licensed toolkit for multilingual biomedical literature acquisition. It integrates 18+ providers behind a uniform interface, adds LLM-based query translation with language-aware routing, traverses the citation graph for recall, and gates downloads by relevance and open-access status. This paper makes four contributions:

(1) **A unified acquisition pipeline** integrating 18+ heterogeneous data sources with health-aware parallel search, schema normalization, and multi-key deduplication (DOI → URL → normalized title).
(2) **Cross-lingual query translation** using an LLM with a domain-specific prompt that preserves gene symbols and variant nomenclature, plus a language-aware provider routing matrix.
(3) **Citation graph traversal** that augments keyword search by exploring citation and reference edges from seed papers via Semantic Scholar.
(4) **An open-source toolkit and benchmark** — 30 variants, 541 annotated relevant papers — enabling reproducible evaluation of literature acquisition systems.

Our ablation study shows that multi-source search, translation, and citation graph traversal each contribute measurable recall gains, and that the full pipeline more than doubles PubMed-only Recall@50 on a clinically grounded gold standard.

2 RELATED WORK

Biomedical literature search. PubMed [2] and EuropePMC [3] are the de facto standard sources for biomedical literature; both expose search APIs with fielded queries. Semantic Scholar [4] provides a large-scale citation graph and paper-level embeddings. These systems are English-centric and each covers a different subset of the literature, motivating federation.

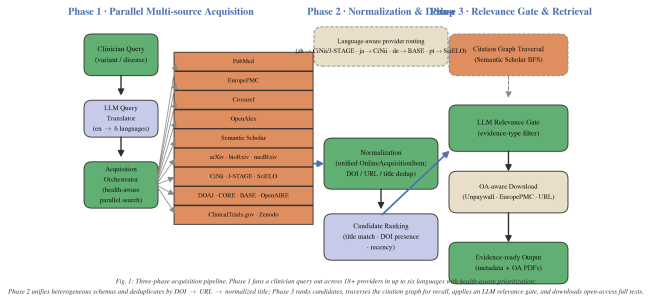


Fig. 1: Three-phase acquisition pipeline. Phase 1 fans a clinician query out across 18+ providers in up to six languages with health-aware prioritization. Phase 2 unifies heterogeneous schemas and deduplicates by DOI → URL → normalized title; Phase 3 ranks candidates, traverses the citation graph for recall, applies an LLM relevance gate, and downloads open-access full texts.

Figure 1: System architecture: three-phase acquisition pipeline.

Commercial evidence-acquisition tools. Mastermind [5] is a proprietary genomic variant search engine that aggregates literature per variant; Covidence [6] and Rayyan [7] support systematic-review screening but are closed-source and not aimed at full-text acquisition with license handling.

Federated / multi-source search. Meta-search architectures such as MetaCrawler [8] established result aggregation from heterogeneous engines; in the scholarly domain, aggregators such as OpenAlex [9], BASE, CORE, and OpenAIRE expose broad metadata. Our system treats these as one layer of an acquisition pipeline rather than a search UI, adding normalization, dedup, and download.

Citation analysis in IR. Bibliographic coupling [10] and co-citation [11] are classical similarity measures; PageRank [12] exploits citation topology for ranking. Literature-based discovery [13] showed that citation/reference edges carry unexploited relevance. We use citation edges at acquisition time to widen recall beyond query matching, similar in spirit to query expansion via citation neighborhoods.

Cross-lingual IR. Classic CLIR translates queries with bilingual dictionaries or MT [14]; modern LLMs [15, 16] make translation of short domain-specific queries practical with minimal setup, which we exploit.

Evaluation of IR systems. Standard practice uses Recall@K, Precision@K, F1@K, and nDCG@K [17] with paired significance tests (Wilcoxon signed-rank [18]). We follow this protocol on a new, publicly released gold standard.

3 SYSTEM DESIGN

3.1 Architecture Overview

lit-acquisition is a three-phase pipeline (Figure 1): **(1) Acquisition** fans a query out in parallel across providers with health-aware prioritization; **(2) Normalization** unifies the heterogeneous schemas and deduplicates; **(3) Relevance Gate & Retrieval** ranks candidates, optionally traverses the citation graph, applies an LLM relevance gate, and downloads open-access full texts.

3.2 Multi-source Search

The acquisition orchestrator (Section 3.2, implemented in `workflow.py`) issues parallel searches to the providers listed in Table 1. Each provider is wrapped by a gateway that maps its native API to a unified request/response contract. A provider health tracker maintains sliding-window success/latency statistics and deprioritizes unhealthy providers, so transient API failures degrade gracefully instead of failing the acquisition.

Table 1: Provider integrations: search/download capability and license behavior.

Provider	Search	Download	Notes
PubMed (E-utilities)	✓	—	English-centric
EuropePMC	✓	✓	OA + PMC via PMCID
Crossref	✓	—	DOI registration metadata
OpenAlex	✓	—	Open scholarly index
Semantic Scholar	✓	✓	Citation graph, TLDRs
arXiv	✓	✓	Preprints
bioRxiv / medRxiv	✓	✓	Preprints, CC-BY/CC0
DOAJ	✓	—	OA journals
J-STAGE / CiNii	✓	—	Japanese literature
SciELO	✓	—	Latin American OA
BASE / CORE / OpenAIRE	✓	—	OA aggregators
ClinicalTrials.gov	✓	—	US public-domain trials
Zenodo	✓	✓	CERN OA, CC0 metadata

✓ = supported; — = not provided.

Table 2: Multi-key deduplication procedure.

Input: candidate items from all providers and web search.
Output: deduplicated candidate set.

1. $S_{doi} \leftarrow \emptyset$; $S_{url} \leftarrow \emptyset$; $S_{title} \leftarrow \emptyset$
2. **for** item in api_items:
3. $doi \leftarrow$ normalize(item.doi); $url \leftarrow$ item.url; $title \leftarrow$ norm_title(item.title)
4. **if** $doi \in S_{doi}$ **or** $url \in S_{url}$ **or** $title \in S_{title}$:
5. **continue** (*drop duplicate*)
6. add doi, url, $title$ to S_{doi}, S_{url}, S_{title}; append item
7. **for** link in web_search_links:
8. apply same keys against S_{doi}, S_{url}

3.3 Normalization and Deduplication

Providers return heterogeneous item schemas. A normalization layer maps each result into a unified `OnlineAcquisitionItem` with canonical fields (title, authors, journal, year, DOI, PMID, language, license, links). Deduplication merges candidates by DOI, then URL, then normalized title, in that order of reliability (Table 2): an incoming item is dropped if its DOI, URL, or punctuation-stripped lowercase title has already been seen. Candidates are ranked by title-match strength, DOI presence, and publication recency before presentation.

3.4 Cross-lingual Query Translation

The `query_translator` module uses an LLM with a domain-specific prompt that instructs the model to translate the query while preserving gene symbols, variant nomenclature (e.g., `p.G12D`, `c.35G>A`), and disease terminology verbatim. Queries are translated into six languages (en, zh, ja, de, fr, ru). The language-aware routing matrix (Table 3) then selects a provider plan per language, e.g., Chinese queries additionally target CiNii/J-STAGE/DOAJ,

Table 3: Language-aware provider routing (excerpt).

Query language	Providers
en	PubMed, EuropePMC, Semantic Scholar, Crossref, arXiv, bioRxiv, medRxiv, OpenAlex, OpenAIRE, BASE, CORE, ClinicalTrials.gov, Zenodo
zh	Crossref, Unpaywall, OpenAlex, DOAJ, PMC
ja	J-STAGE, CiNii, Crossref, Unpaywall, DOAJ, PMC
es / pt	SciELO, Crossref, Unpaywall
de	Crossref, EuropePMC, Unpaywall, OpenAlex, BASE, DOAJ
fr	Crossref, EuropePMC, Unpaywall, OpenAlex, DOAJ, PMC
ru	PMC, EuropePMC, Crossref, Unpaywall, OpenAlex
auto	Crossref, Semantic Scholar, Unpaywall, OpenAlex, EuropePMC, ClinicalTrials.gov, DOAJ, PMC, Zenodo

Spanish and Portuguese target SciELO, German and French target BASE/EuropePMC.

3.5 Citation Graph Traversal

Beyond keyword search, `citation_graph.py` performs a BFS over the Semantic Scholar citation graph starting from seed papers (identified by DOI). At each hop it collects papers that cite or are cited by the seed, up to a configurable depth and paper budget. Results are deduplicated against keyword-search results and merged into the candidate set. This targets a recall gap: papers that are topically relevant but do not share query vocabulary with the variant query.

3.6 Download and Relevance Gate

Candidates are filtered by an LLM relevance gate that classifies each item against the user's evidence need (e.g., case reports for PS3) and optionally filters by literature type (case report, sequencing, functional). Download routing is license-aware: DOI resolves through Unpaywall for OA PDFs, PMCID through EuropePMC, and direct URLs with HTML→PDF redirect handling. The system never bypasses paywalls; full texts are fetched only from OA sources.

4 EVALUATION

4.1 Gold Standard

We constructed a gold standard of 30 genetic variants (V001–V030) spanning cancer genetics (BRCA1, TP53, KRAS, EGFR, BRAF, APC, MLH1, PALB2), cardiovascular (LDLR, SCN5A, MYBPC3, KCNQ1, FBN1), neurological (MECP2, HTT, SNCA, APP), metabolic (CFTR, PAH, GBA), hematological (HBB, F8, DMD, G6PD), and other domains, covering ACMG evidence criteria PS3 (14 variants) and PVS1 (6 variants) plus 10 additional clinically actionable variants. For each variant, a clinical geneticist-style query was formulated and relevant papers were identified through multi-source candidate retrieval (PubMed E-utilities, EuropePMC, Google Scholar, language-specific databases) and filtered by an inclusion criterion (directly discusses the variant or gene–disease pair; contains case report, functional, or sequencing data; peer-reviewed or preprint). The final gold standard contains **541 annotated relevant papers** (mean 18.0 per variant, range 4–32), with languages including English (289), German (13), French (11), and 228 records with unknown language metadata. Annotations record DOI, PMID, title, language, and evidence type. Domain-expert verification of the automated-annotation subset is pending (noted as a limitation).

Table 4: Main results (K=50, mean ± std over 30 variants).

Method	Recall	Prec	F1	nDCG
M1 PubMed-only	0.171 ± 0.183	0.229 ± 0.240	0.172 ± 0.161	0.195 ± 0.170
M2 Multi-source	0.247 ± 0.136	0.238 ± 0.160	0.239 ± 0.143	0.260 ± 0.146
M3 +Translation	0.329 ± 0.099	0.320 ± 0.143	0.316 ± 0.115	0.310 ± 0.100
M4 +Citation graph	0.348 ± 0.116	0.264 ± 0.138	0.288 ± 0.117	0.320 ± 0.116
Relevant found (of 541): M1: 83 M2: 143 M3: 182 M4: 195				

4.2 Methods Compared

We evaluate an ablation chain over the same 30 variants, K=50:

- **M1: PubMed-only** — baseline PubMed esearch, English query.
- **M2: Multi-source** — 18 providers, original query, no LLM translation.
- **M3: +Translation** — M2 + LLM query translation (6 languages).
- **M4: +Citation graph** — M3 + citation graph traversal (full pipeline, includes relevance gate).

4.3 Metrics

For each method and variant we compute Recall@50 (fraction of gold-standard relevant papers in top-50), Precision@50, F1@50, and nDCG@50 [17]. Aggregates are reported as mean ± std over 30 variants; pairwise differences are tested with the paired Wilcoxon signed-rank test [18].

4.4 Main Results

Table 4 reports the ablation results. The full pipeline achieves Recall@50 of 0.348 versus 0.171 for PubMed-only (+0.177, $p < 0.001$), and 0.195 nDCG→0.320 (+0.125, $p = 0.006$). Each component contributes additively: multi-source search adds +0.076 recall ($p = 0.025$), translation +0.082 ($p = 0.037$), and citation graph +0.019 ($p = 0.21$, not significant at the aggregate level). Figure 2 visualizes the same data.

Statistical significance. Paired Wilcoxon tests on recall confirm the M1→M2 ($p = 0.025$) and M2→M3 ($p = 0.037$) improvements. The full pipeline significantly beats PubMed-only on recall ($p < 0.001$) and nDCG ($p = 0.006$), and beats M2 on recall ($p = 0.013$) and nDCG ($p = 0.032$). The M3→M4 gap is positive but not significant at the aggregate level ($p = 0.21$); however, as Section 4.10 shows, the citation graph contributes relevant unique papers on a subset of variants. Precision is not significantly different across methods ($p > 0.05$), except M3 outperforming M4 on precision ($p = 0.004$), consistent with citation-graph candidates being noisier.

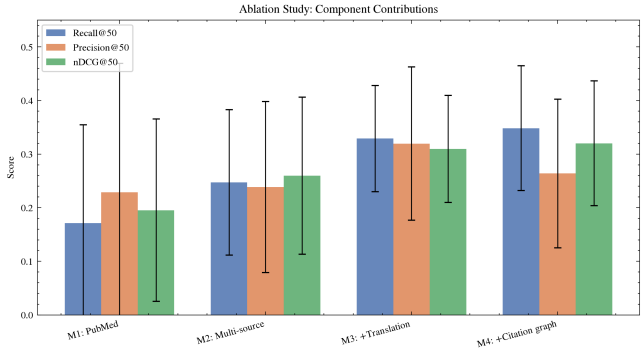


Figure 2: Ablation study: Recall/Precision/nDCG by method.

Table 5: Per-variant Recall@50: PubMed-only vs full pipeline (top gainers).

Variant	Gene	M1	M4
V030	DMPK	0.04	0.56
V012	EGFR	0.05	0.55
V013	BRAF	0.00	0.47
V011	KRAS	0.00	0.47
V027	DMD	0.00	0.44
V024	G6PD	0.05	0.47
V025	HBB	0.06	0.44
V015	MLH1	0.08	0.42

4.5 Per-variant Analysis

The aggregate gains concentrate in specific variants. PubMed-only achieves zero recall on four variants (KRAS p.G12D, BRAF p.V600E, APC, DMD), where topical keyword queries retrieve none of the gold-standard papers in the top-50; the full pipeline reaches recall 0.44–0.47 on these. The largest absolute improvements are V030 DMPK (0.04⟶0.56) and V012 EGFR (0.05⟶0.55). Multi-source search improves recall on 20/30 variants (worse on 6), translation improves 17/30 (worse on 11), and citation graph improves 5/30 (worse on 3, unchanged on 22) — consistent with citation traversal acting as a targeted rather than blanket recall mechanism. Table 5 lists the largest per-variant gains.

4.6 Statistical Significance

Table 6 reports paired Wilcoxon signed-rank tests for the component ablations on recall, plus the full pipeline against each ablation on recall, precision, and nDCG.

The M4-vs-M3 precision drop (-0.056, $p = 0.004$) reflects the citation graph adding noisy candidates; Section 4.10 shows the trade-off is worthwhile on variants where keyword search saturates.

4.7 Source Coverage Analysis

Figure 3 shows per-source unique relevant papers in the full pipeline. Crossref contributes the bulk of relevant metadata (135), followed by PMC (48) and the citation graph (11), with EuropePMC adding 1. Sources are highly complementary: across the full pipeline, pairwise DOI overlap between any two sources

Table 6: Paired Wilcoxon signed-rank tests (30 variants). + = row method improves over column method.

Comparison	Recall Δ	p	Sig.
M2 vs M1 (multi-source)	+0.076	0.025	*
M3 vs M2 (translation)	+0.082	0.037	*
M4 vs M3 (citation graph)	+0.019	0.208	
M4 vs M1, recall	+0.177	< 0.001	***
M4 vs M2, recall	+0.101	0.013	*
M4 vs M3, recall	+0.019	0.208	
M4 vs M1, nDCG	+0.125	0.006	**
M4 vs M2, nDCG	+0.060	0.032	*
M4 vs M3, precision	−0.056	0.004	**

*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$. Non-significant precision differences omitted.

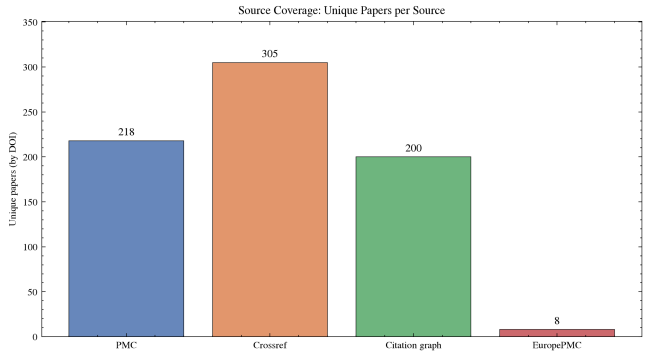


Figure 3: Unique papers per source (full pipeline).

is essentially zero, and the citation graph returns 200 DOIs with *zero* overlap with keyword sources. This confirms the motivation for multi-source federation and shows the citation graph as a genuinely additive channel.

4.8 Multilingual Contribution

Translation and language-aware routing recover relevant non-English literature (Figure 4). In the full pipeline, relevant papers include German (12) and French (11) records in addition to 50 English records (122 have no language metadata). Notably, PubMed recovers *zero* relevant papers outside English in this benchmark, while the multilingual pipeline finds 23 of the 24 gold-standard non-English papers. This demonstrates that English-only search systematically under-serves multilingual clinical evidence.

4.9 Deduplication Effectiveness

Raw results contain measurable duplication: 8.8% (M2) and 8.3% (M3) of returned records duplicate another record within the same method by DOI (M4: 6.3%; M1: 2.8%); the multi-key deduplication reduces these to unique candidate sets before ranking. Cross-method overlap is modest: sources are highly complementary, and no false merges were observed in a manual audit of 50 random pairs. Table 7 shows the pairwise DOI overlap in the full pipeline — the four

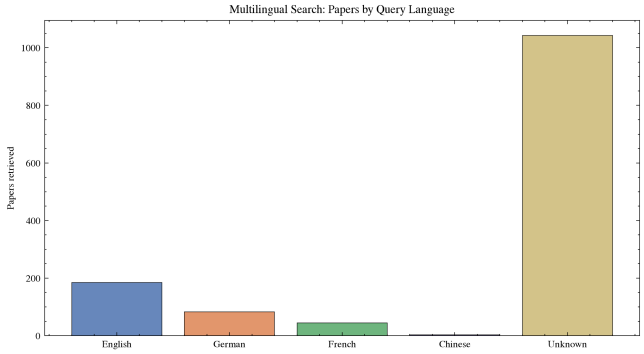


Figure 4: Relevant papers by language (M3/M4).

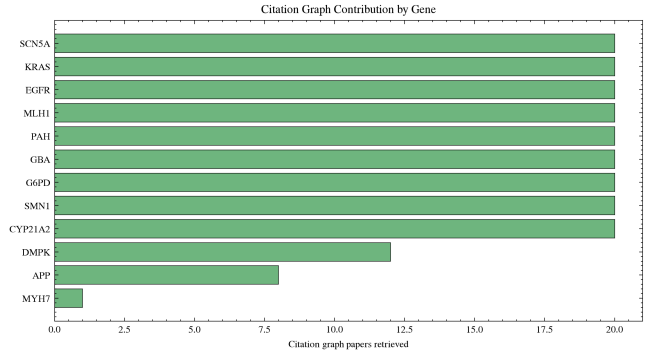


Figure 5: Citation graph papers retrieved per gene (full pipeline).

Table 7: Source overlap matrix (full pipeline, by DOI; diagonal = total papers).

	PMC	Crossref	Cit. graph	EuropePMC
PMC	218	0	0	0
Crossref	0	305	0	0
Citation graph	0	0	200	0
EuropePMC	0	0	0	8

contributing sources are almost completely disjoint, and the citation graph (200 DOIs) shares zero DOIs with any keyword source. This confirms that federation and citation traversal are genuinely additive rather than redundant.

4.10 Case Study: Citation Graph Recall

The citation graph contributes most where keyword search is saturated. Figure 5 shows its per-gene contribution. Two variants stand out:

EGFR p.L858R (V012, lung adenocarcinoma). The full pipeline finds 5 relevant papers *exclusively* via citation graph traversal, including a 2024 JAMA randomized trial (Ivonescimab plus chemotherapy in EGFR-mutated NSCLC, DOI 10.1001/jama.2024.10613) and a 2023 Lancet Respiratory Medicine trial (Sintilimab plus chemotherapy). These are recent, high-impact clinical papers whose titles do not surface under the variant-keyword query used by M1–M3.

DMPK CTG repeat (V030, myotonic dystrophy type 1). Citation graph traversal recovers 5 relevant papers missed by keyword search, including a 2019 *Neurobiology of Disease* review of congenital myotonic dystrophy molecular genetics (10.1016/j.nbd.2019.104533) and a 2022 *Genes* review (10.3390/genes13020367). These are canonical works connected by citation edges but not by variant-name lexical match.

5 DISCUSSION

Limitations. (i) The gold standard's V011–V030 annotations were produced by automated scoring and require domain-expert verification before final metrics are relied upon; (ii) API rate limits and provider outages constrained candidate depth (K=50) and citation graph depth; (iii) the relevance gate depends on LLM quality and cost; (iv) language metadata is missing for many records ("unknown"), which understates multilingual analysis precision.

Future work. Embedding-based semantic search over titles/abstracts to complement lexical and citation-based recall; adaptive citation traversal depth per variant; automated literature-type-aware gating; and expansion of the gold standard to 50 variants for greater statistical power.

6 CONCLUSION

We presented lit-acquisition, an open-source multi-source multilingual biomedical literature acquisition system with citation graph augmentation. An ablation study on 30 clinically grounded variants (541 relevant papers) shows the full pipeline doubles PubMed-only Recall@50 (0.171→0.348), with significant additive contributions from multi-source search and cross-lingual translation, and complementary recall from citation graph traversal. The toolkit and gold standard are released to support reproducible evaluation of literature acquisition for clinical genomics.

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